MARMA-CRYPTO & DSTU 4145-2002 GEOMETRIC SIGNATURE PROTOCOL

DSTU 4145-2002 Elliptic Curve GF(2^257) · GOST 34.311-95 · DSTU 7564 · 256-Bit Marma Audit

Timestamp (UTC): 2026-10-24 17:41:37 UTC

VERIFIED: TRUE · DSTU 4145-2002 ELLIPTIC CURVE & 256-BIT SRI YANTRA GEOMETRY ARE VALID

Target File:	00_sample_marma_signed.pdf
Crypto Standard:	···· 4145-2002 (··· ·····) / ···· ···· 34.311-95 / ···· 7564:2014
Signer Subject (CN / RNOKPP):	····· ····· ····· (RNOKPP: 3255419579)
Certificate Authority (QTSP):	····· monobank | Universal Bank
Key Medium & Fingerprint:	·····_·_·_mono···_2025-09-15T20_12_28.pfx [b5913a2f269eb107db3ecf30...]
GOST 34.311-95 Digest:	5e55e114c8dddc98528b4c74ae6e2cf56a8f58abe46c4f9777f0bdcf3849ce75
DSTU 7564-256 (Kupyna):	9226064d8c272adf0bbf125f168a784f6ea61f168a345e058401bc6ec68bfcfd
Parallel Combiner SHA-256:	0fab57cc232056e03d16dfa273f4a13768d376a5bae8ae6bebb6d69e0995b628
DSTU 4145-2002 Vector (r):	0000000049aa4e40667a89cd94a183692752d694691820e0432c8af65f5c056b
DSTU 4145-2002 Vector (s):	00000000475934d419f032336753b2257986aca73308917a81da41a87e4f8bf3
Max Coordinate Drift (·_coord):	0.0000000000 (threshold < 1e-8)
Bilateral Symmetry Error (·_sym):	0.000000000000 (invariant lx + rx = 300.0)

9-Triangle Micro-Perturbation Coordinate Audit Table (Bound to DSTU 4145-2002 Signature Vector r, s):

Triangle	left_x (lx)	base_y (by)	apex_y (ay)	right_x (rx)	lx + rx (Sym)
D1	53.656698471	123.205083397	250.000002565	246.343301529	300.000000000
U1	52.984006377	174.246608247	50.000001593	247.015993623	300.000000000
U3	98.718237387	220.038281307	123.205077889	201.281762613	300.000000000
U2	78.264677487	197.923164453	78.104994759	221.735322513	300.000000000
D3	90.485687241	78.105000123	160.660146763	209.514312759	300.000000000
D2	80.983848309	103.121998389	220.038294411	219.016151691	300.000000000
U4	114.948847657	160.660153549	103.121996139	185.051152343	300.000000000
D4	116.351419903	134.307582399	197.923153833	183.648580097	300.000000000
D5	124.611916877	144.797772927	174.246605610	175.388083123	300.000000000